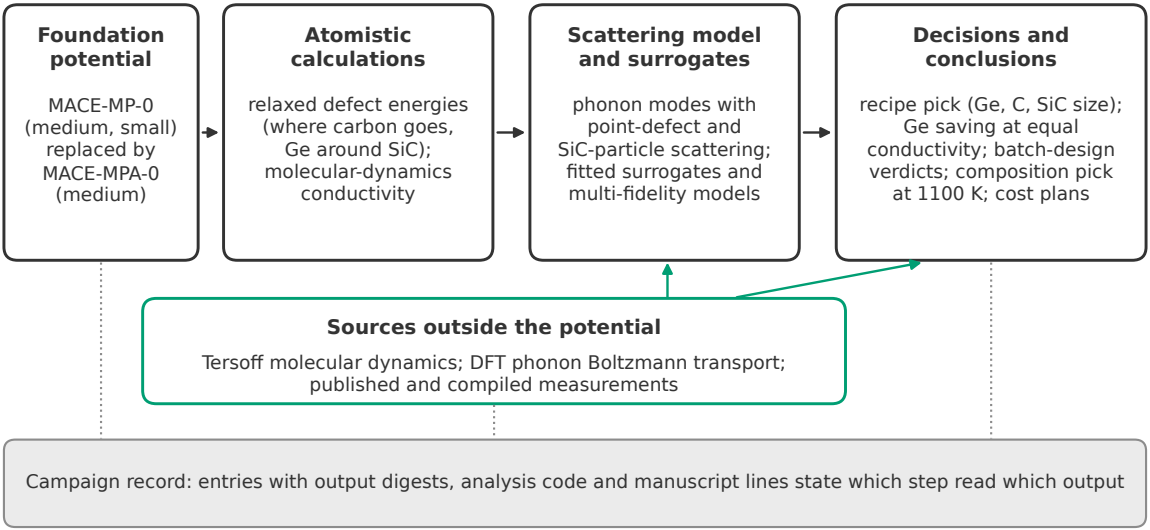
Foundation potential
MACE-MP-0 (medium, small) replaced by MACE-MPA-0 (medium)
Atomistic calculations
relaxed defect energies (where carbon goes, Ge around SiC); molecular-dynamics conductivity
Scattering model and surrogates
phonon modes with point-defect and SiC-particle scattering; fitted surrogates and multi-fidelity models
Decisions and conclusions
recipe pick (Ge, C, SiC size); Ge saving at equal conductivity; batch-design verdicts; composition pick at 1100 K; cost plans
Sources outside the potential
Tersoff molecular dynamics; DFT phonon Boltzmann transport; published and compiled measurements
Campaign record: entries with output digests, analysis code and manuscript lines state which step read which output